```matlab
%
===========================================================================
% Simulation of Stepper Motor Operation
% Speed and Position Control
%
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clear; clc; close all;

%% 1. Define Motor Parameters
step_angle = 15; % Degrees per step (Common for NEMA 17 motors)
steps_per_rev = 360 / step_angle; % Total steps for one full revolution
(200)

%% 2. User Inputs for Control
target_position_deg = 1080; % Desired position in degrees (e.g., 720 = 2
revolutions)
pulse_frequency = 250;     % Speed: Pulses per second (Hz)

% Calculate the exact number of discrete steps needed
total_steps_required = round(target_position_deg / step_angle);

% Calculate the time duration of a single step
time_per_step = 1 / pulse_frequency;

% Total simulation time
total_time = total_steps_required * time_per_step;

%% 3. Simulation Arrays
% Create a time vector mapping each discrete step
t = 0 : time_per_step : total_time;

% Initialize position and speed arrays
position = zeros(1, length(t));
speed_rpm = zeros(1, length(t));

%% 4. Run Step-by-Step Simulation
for i = 2:length(t)
    % Position increases discretely by the step angle
    position(i) = position(i-1) + step_angle;

    % Speed calculation:
    % (Steps/sec) * (Degrees/step) = Degrees/sec. Convert to RPM.
    speed_rpm(i) = (pulse_frequency * step_angle) * (60 / 360);
end

%% 5. Plotting the Results
figure('Name', 'Stepper Motor Simulation', 'NumberTitle', 'off',
'Position', [100, 100, 800, 600]);

% Plot 1: Position Control (Staircase plot to show discrete steps)
subplot(2,1,1);
stairs(t, position, 'b-', 'LineWidth', 2);
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title('Position Control: Motor Angle vs Time');
xlabel('Time (Seconds)');
ylabel('Position (Degrees)');
grid on;
hold on;
% Mark the final target position
plot(t(end), position(end), 'ro', 'MarkerFaceColor', 'r');
text(t(end)*0.8, position(end)*0.9, sprintf(' Target: %d°',
target_position_deg));

% Plot 2: Speed Control
subplot(2,1,2);
plot(t, speed_rpm, 'r-', 'LineWidth', 2);
title('Speed Control: Motor Speed vs Time');
xlabel('Time (Seconds)');
ylabel('Speed (RPM)');
ylim([0, max(speed_rpm) * 1.5]); % Scale Y-axis for better visibility
grid on;
```